

8.2.2 Electric field of point charge:

To find the electric field due to a charged particle, consider a point charge q as a source charge. This charge creates an electric field at all points in space surrounding it. A test charge q_0 is placed at a distance r from the source charge. (see figure 8.6).

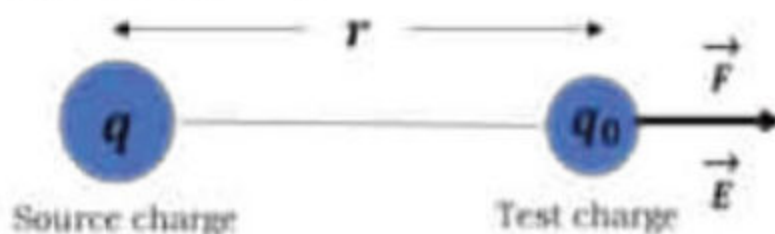


Fig: 8.6

A test charge q_0 in the field of source charge q .

The test charge experiences electric force from which we can determine the direction of the electric field. According to Coulomb's law, the force exerted by q on the test charge is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r}$$

where $\hat{r}$ is a unit vector directed from q toward q_0 and the direction of force $\vec{F}$ is directly away from the source charge q . To calculate the electric field at a distance r from the point charges, we first calculate the electric field vector using equation 8.3.

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (8.4)$$

The direction of electric field directly away from the point charge q as shown in figure 8.9.

The magnitude of electric field at any given distance r can expressed as:

$$\begin{aligned} \vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \\ \vec{E} &\propto \frac{1}{r^2} \end{aligned}$$

This equation shows that the intensity of the electric field decreases with distance from the source charge(s). The field intensity is inversely proportional to the square of the distance from the charge(s).

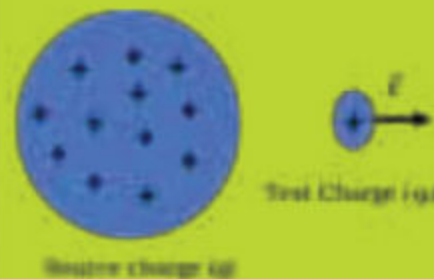
If there are n number of charged particles $q_1, q_2, q_3, \dots, q_n$, then the net electric force of these charges at a point. According to superposition principle,

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n$$

From equation 8.6, the net electric field of these charged particles

DO YOU KNOW?

A small positive test charge q_0 placed near a source charge carrying a much greater positive charges q as shown in figure. The test charge is always so small that the field of the source charge is unaffected by its presence.



The electric field strength due to the source charge at the location of the test charge is defined as the electric force on the test charge per unit charge.

$$\vec{E} = \frac{\vec{F}_1}{q_0} + \frac{\vec{F}_2}{q_0} + \frac{\vec{F}_3}{q_0} + \dots + \frac{\vec{F}_n}{q_0}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots + \vec{E}_n$$

Therefore, the electric fields of these charges experienced by the test charge q_0 is vector sum of electric field of individual charge.

Worked Example 8.4

Find the intensity of electric field at a point such that a proton placed at it would experience a force equal to its weight.

Solution:

Step:1 Write down the known quantities and quantities to be found.

Mass of proton = $m_p = 1.672 \times 10^{-27} \text{ kg}$,

Charge on proton = $q_p = 1.602 \times 10^{-19} \text{ C}$ and $g = 9.8 \text{ m/s}^2$

Step:2 Write down the formula and rearrange if necessary.

$$\vec{E} = \frac{F}{q_p}$$

as the electric force equal to its weight, i.e. $F = W = mg$,

$$\vec{E} = \frac{mg}{q_p}$$

Step:3 Put the values in formula and calculate.

$$\vec{E} = \frac{(1.672 \times 10^{-27} \text{ kg}) \times 9.8 \text{ m/s}^2}{1.602 \times 10^{-19} \text{ C}}$$

$$\vec{E} = 10.22 \times 10^{-8} \text{ N/C}$$

Worked Example 8.5

- Calculate the magnitude of electric field strength, if a test charge $q_0 = +3.5 \mu\text{C}$ experience a force of 70 mN (field lines of electric field shown in figure A).
- if this test charge is replaced by an electron, then calculate force on an electron and state the direction of force.

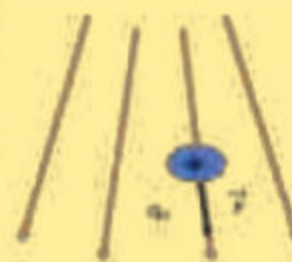


Figure A

Solution:

Step:1 Write down the known quantities and quantities to be found.

$F = 70 \text{ mN} = 70 \times 10^{-3} \text{ N}$, $q_0 = 3.5 \mu\text{C} = 3.5 \times 10^{-6} \text{ C}$

Charge on electron = $q_e = 1.602 \times 10^{-19} \text{ C}$

(a) $E = ?$

(b) $F = ?$

Step:2 Write down the formula and rearrange if necessary.

$$E = \frac{F}{q_0} \text{ and } F = Eq_e$$

Step:3 Put the values in formula and calculate.

$$(a) E = \frac{70 \times 10^{-3} \text{ N}}{3.5 \times 10^{-6} \text{ C}} = 2.0 \times 10^4 \text{ N/C}$$

$$(b) F = 2.0 \times 10^4 \text{ N/C} \times 1.602 \times 10^{-19} \text{ C}$$

$$F = 3.204 \times 10^{-15} \text{ N}$$

The magnitude of the force experienced by an electron is $3.204 \times 10^{-15} \text{ N}$.

The direction of the force on an electron is directly upwards because field lines are directly downwards and the charge on an electron is negative.

Self-Assessment Question:

A certain gas is filled in a tube and when the electric field reaches to a certain maximum value across the tube then the gas becomes conducting. Explain why the gas becomes conducting.

8.3.1 Electric Dipole:

An electric dipole is a simple system in electromagnetism consisting of two opposite electric charges of equal magnitude, separated by a small distance d as shown in figure. The charges create an electric field that has a distinct pattern, with field lines oriented along the axis of the dipole. The strength and direction of the dipole are represented by its dipole moment. The dipole moment $\vec{p}$ is a vector quantity that represents the strength and direction of the electric dipole. It is defined as the product of the charge magnitude (q) and the distance between the charges d :

$$\vec{p} = q \times d$$

Electric dipoles play a fundamental role in understanding the behavior of polar molecules, interactions in electric fields, and various electrical phenomena in physics and engineering.

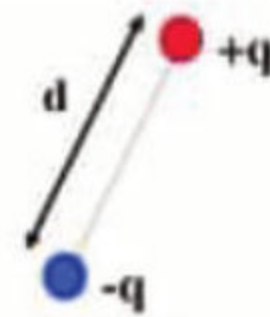


Fig: 8.7

8.3.2 Electric field at point due to two charges:

A pair of equal and opposite point charges separated by a small distance form an electric dipole. To calculate the electric field of the dipole at point C which is at a distance “y” from the center of dipole. Consider two charges “-q” and “+q” placed at a small distance “d” from each other as shown in figure 8.8. The charge “+q” sets up field E_+ and the charge “-q” produces the field E_- (see the figure for the direction of the fields). Resolving the field vectors into the components. From figure, it is clear that the vertical components “ $E_+ \cos\theta$ ” and “ $E_- \cos\theta$ ” cancel each other. Therefore, the net electric field is due to the vector sum of the horizontal components.

$$\vec{E} = E_+ \cos\theta + E_- \cos\theta$$

The magnitude of both electric fields is the same. From equation (8.4), we have

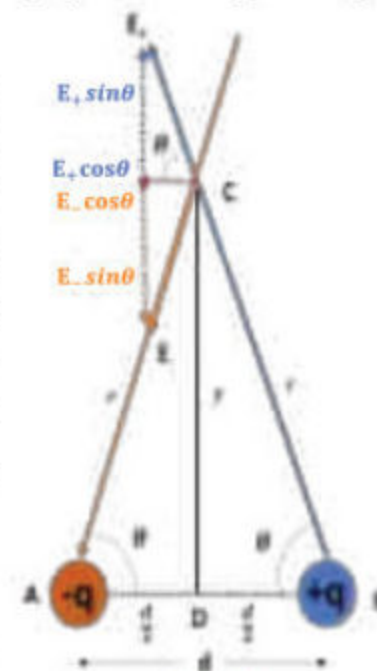


Fig: 8.8

Electric dipole formed by two charges +q and -q.

$$E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$E = 2 \times \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \cos\theta$$

From figure, In triangle ADC,

$$\cos\theta = \frac{d/2}{r} = \frac{d}{2r}$$

Putting the value of $\cos\theta$, we get.

$$E = 2 \times \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \frac{d}{2r}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{r^3}$$

From figure, In triangle ADC,

$$r = \sqrt{y^2 + (d/2)^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{\left(y^2 + (d/2)^2\right)^{3/2}}$$

where the product of the “ q ” and “ d ” is known as the electric dipole moment “ P ” of the dipole. The electric dipole moment is a vector quantity and it is always directed from the negative to the positive charge.

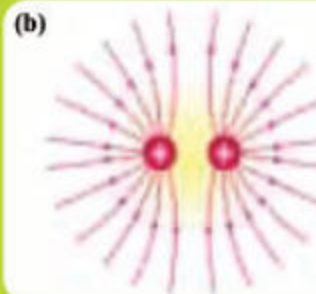
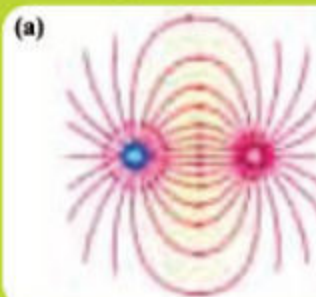
$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3 \left(1 + (d/2y)^2\right)^{3/2}}$$

As $d \ll y$, therefore approximate electric field intensity of the dipole can be calculated by neglecting the term $(d/2y)^2$.

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3} \quad (8.5)$$

DO YOU KNOW?

Do you know about the electric field lines of two point charges?



Worked example 8.6

Calculate the electric field intensity at point P of two positive charges of same magnitude which are separated by small distance " d " as given in figure.

Solution:

Resolving the field vectors into the components. From figure, it clear that the horizontal components cancel each other. Therefore, the net electric is due to the vector sum of the vertical components.

$$E_{\text{total}} = E \sin\theta + E \sin\theta$$

$$E_{\text{total}} = 2E \sin\theta$$

where $\sin\theta = \frac{y}{r}$ and using equation (8.7), we get

$$E = 2 \times \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \frac{y}{r}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{qy}{r^3}$$

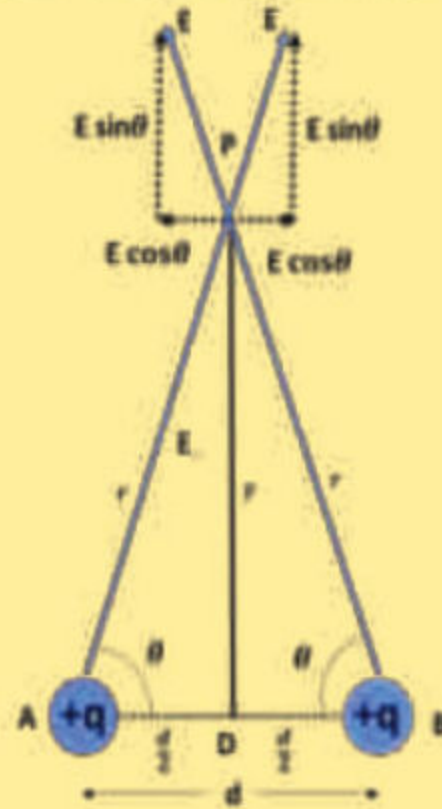
where $r = \sqrt{y^2 + (d/2)^2}$

$$E = \frac{1}{2\pi\epsilon_0} \frac{qy}{(y^2 + (d/2)^2)^{3/2}}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{qy}{y^3 (1 + (d/2y)^2)^{3/2}}$$

As $d \ll y$, therefore approximate electric field intensity due to two positive charges can be calculated by neglecting the term $(d/2y)^2$.

$$E = \frac{1}{2\pi\epsilon_0} \frac{q}{y^2}$$

**8.4.1 Electric Flux:**

To understand the electric field lines quantitatively, we consider electric field that is uniform in both magnitude and direction. The field lines penetrate a rectangular surface of area A , whose plane is oriented perpendicular to the field E as shown in **figure 8.9**. The total number of lines penetrating the surface is proportional to the dot product of EA . Therefore, the dot product of the electric field $\vec{E}$ and vector area $\vec{A}$ is called the electric flux and denoted by Φ_e .

$$\Phi_e = \vec{E} \cdot \vec{A} \quad (8.6)$$

where vector area A is parallel to electric field lines. The SI unit of electric flux is $N.m^2/C$.

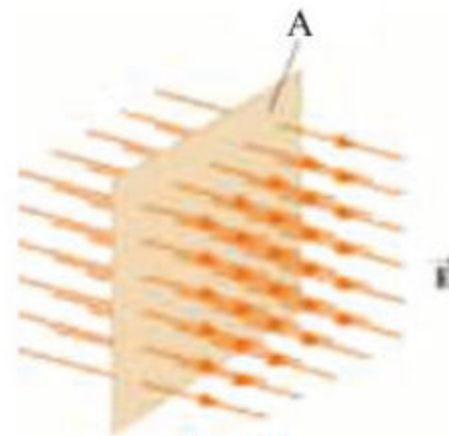


Fig: 8.9

The field lines penetrating through a rectangular surface of area.

If the electric field line penetrating through a surface in such a way that the vector area A is parallel to the field, then the maximum field lines will penetrate through the surface. From above equation,

$$\Phi_e = EA \cos 0^\circ = EA$$

If the electric field line penetrating through a surface in such way that the vector area A is perpendicular to the field, then no field lines will penetrate through the surface.

$$\Phi_e = EA \cos 90^\circ = 0$$

If the electric field line penetrating through a surface in such way that the vector area A is antiparallel to the field, then the electric flux will be negative.

$$\Phi_e = EA \cos 180^\circ = -EA$$

8.4.2 Electric Flux through a surface enclosing a charge:

The definition of electric flux given by equation (8.6) is valid for a small element of area over which the field is approximately constant. To calculate the electric flux through a surface enclosing a charge. This surface is divided into a N number of small elements of area $\Delta A_1, \Delta A_2, \Delta A_3, \dots, \Delta A_N$ as shown in figure 8.10. According to the equation (8.9), the values electric flux through these small area elements are $\vec{E} \cdot \Delta \vec{A}_1, \vec{E} \cdot \Delta \vec{A}_2, \vec{E} \cdot \Delta \vec{A}_3, \dots, \vec{E} \cdot \Delta \vec{A}_N$. Therefore, the net electric flux through the closed surface is given as:

$$\Phi_e = \vec{E} \cdot \Delta \vec{A}_1 + \vec{E} \cdot \Delta \vec{A}_2 + \vec{E} \cdot \Delta \vec{A}_3 + \dots + \vec{E} \cdot \Delta \vec{A}_N$$

$$\Phi_e = \sum_{i=1}^N \vec{E} \cdot \Delta \vec{A}_i$$

We are often interested to evaluate the flux through a closed surface such as the flux through a surface of a sphere. The electric flux through a sphere can be calculated by considering a sphere of radius " r " enclosing the charge " q ". If we divide the whole sphere into small area element, then the field lines are penetrating parallel to the vector area for each area element. The area of the sphere is $\sum_{i=1}^N \Delta A_i = 4\pi r^2$ and electric field at the surface of sphere is $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$. From equation (8.10), the net electric flux through this sphere is given as

$$\Phi_e = \frac{q}{\epsilon_0}$$

This equation tells us that the net electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ time the total charge enclosed in that surface which is known as Gauss's Law.

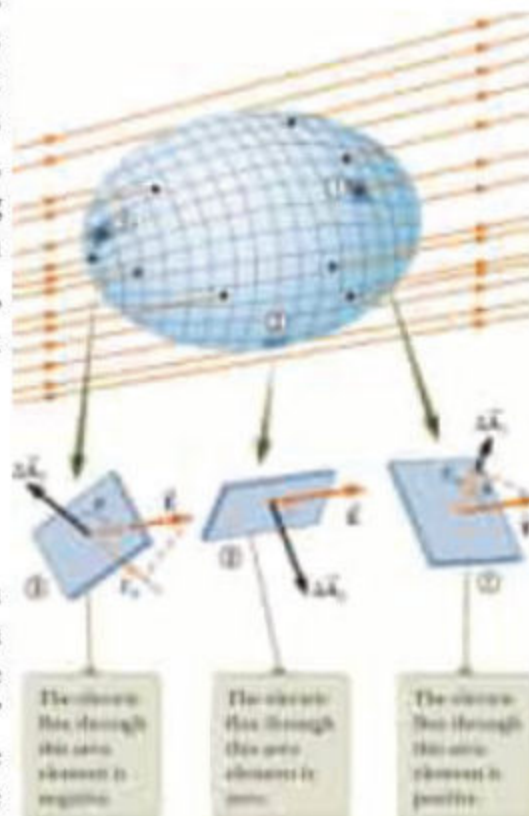


Fig: 8.10

The electric lines of force produced by the source charge enclosed in the closed surface.

Self-Assessment Questions:

1. Sketch the electric field lines for two equal and opposite point charges placed near each other.
2. Calculate the flux of a uniform electric field strength $E = 6\hat{i} \text{ N/C}$ through surface of vector area $A = (3\hat{i} + 5\hat{j}) \text{ m}^2$.
3. A plane surface is rotated in a uniform electric field. When is the flux of the electric field through the surface maximum? Explain with diagram.

8.5.1 Potential difference:

When a test charge " q_0 " is placed in an electric field " E " created by the source charge then this test charge experiences some force (q_0E). This work is done by some external agent to displace the test charge through a distance " r " from one point to another point against the electric field. This work is stored in form of electric potential energy (U), i.e.,

$$W = U = Fr = q_0Er$$

This situation is analogous to that of lifting an object with mass in a gravitational field: the work done by the external agent is mgh as discussed in Chapter 5. The electric potential is defined as the work done by the test charge to move it from lower potential to the higher potential and given as.

$$V = \frac{W}{q_0} = \frac{U}{q_0} \quad (8.7)$$

Worked Example 8.7

An alpha particle ($+2e$) in a nuclear accelerator moves from one terminal at the potential of $6.5 \times 10^6 \text{ V}$ to another terminal at zero potential. What is the corresponding change in the potential energy of the system?

Solution:

Step:1 Write down the known quantities and quantities to be found.

Charge on alpha particle $q = 2e = 2 \times 1.6 \times 10^{-19} \text{ C}$

The electric potential at one terminal $= V_1 = 6.5 \times 10^6 \text{ V}$

The electric potential at other terminal $= V_2 = 0 \text{ V}$

$$\Delta U = ?$$

Step:2 Write down the formula and rearrange if necessary.

From equation (8.11), we have $V_1 = \frac{U_1}{q}$, $V_2 = \frac{U_2}{q}$,

Therefore, the change in potential energy

$$\Delta U = q(V_2 - V_1)$$

Step:3 Put the values in formula and calculate.

$$\Delta U = 2 \times 1.6 \times 10^{-19} \text{ C} (0 - 6.5 \times 10^6 \text{ V})$$

$$\Delta U = -2.1 \times 10^{-12} \text{ J}$$

Hence, the potential energy of the system is $2.1 \times 10^{-12} \text{ J}$.

Worked Example 8.8

For what value of change in electric potential to remove the six electrons from the carbon atom if the change in potential energy is 120 keV.

Solution:

Step:1 Write down the known quantities and quantities to be found.

$$q = 6e = 6 \times 1.6 \times 10^{-19} \text{ C}$$

$$\Delta U = 120 \text{ keV} = 120 \times 1.6 \times 10^{-19} \text{ kV}$$

Step:2 Write down the formula and rearrange if necessary.

$$\Delta V = \frac{\Delta U}{q}$$

Step:3 Put the values in formula and calculate.

$$\Delta V = \frac{120 \times 1.6 \times 10^{-19} \text{ kV}}{6 \times 1.6 \times 10^{-19} \text{ C}}$$

$$\Delta V = 20 \text{ kV}$$

Hence, the value of change in electric potential is 20 kV.

8.5.2 Electric Potential of point charge:

The electric potential can be calculated by considering a point charge q as a source charge. This charge creates an electric field at all points in space surrounding it. A test charge q_0 is placed at a distance r from the point charge as shown in figure 8.11. Putting the value of work done ($W = q_0 E r$) in equation (8.7), we get $V = E r$.

Substituting the value of electric field of point charge from equation (8.4), we get

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Figure 8.16 shows that the potentials at two different places in the field of the charge " q ".

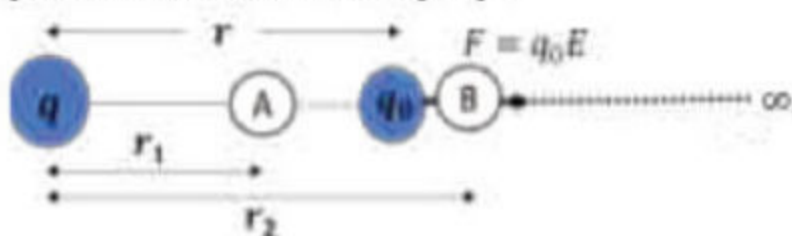


Fig. 8.11: A test charge q_0 in the field of source charge q .

This potential difference can be written as:

$$V_2 - V_1 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$$

where V_1 and V_2 are electric potentials at a distance r_1 and r_2 respectively.

From above relation, the absolute potential is the work done on a unit charge to bring it from infinity to a certain point in the electric field.

DO YOU KNOW?

This relation can also be used to measure the Hall potential with the help of voltmeter during the Hall Effect experiment.

DO YOU KNOW?

Sharks have special organs that are very sensitive to electric field and can detect potential difference of the order of nano-volt and can locate their prey very precisely.



8.5.3 Calculating electric field from electric potential:

Suppose that a positive test charge q_0 moves through a displacement from one-point A to another point B against the electric field as shown in figure 8.12. The work done in moving the test charge " q_0 " through a displacement Δr against the field.

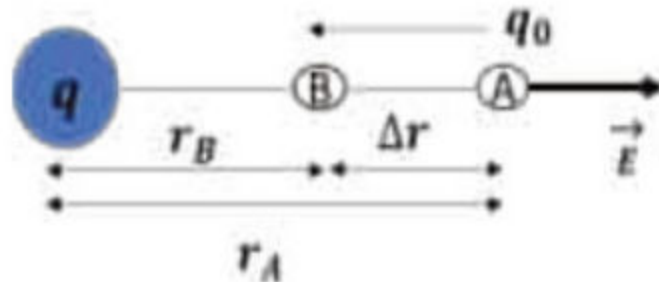


Figure: 8.12: The potentials difference between two points in the electric field.

$$\Delta W = -F\Delta r = -q_0 E \Delta r$$

From the definition of electric potential,

$$\text{we have } \Delta V = \frac{\Delta W}{q_0} = \frac{-q_0 E \Delta r}{q_0}$$

$$E = -\frac{\Delta V}{\Delta r} \quad (8.8)$$

where $\frac{\Delta V}{\Delta r}$ represent the maximum rate of change of potential with respect to the distance known as potential gradient. Thus, the above equation shows that the electric field intensity can be calculated from the negative potential gradient.

DO YOU KNOW?

A unit of energy commonly used in atomic and nuclear physics is the electron volt (eV), which is defined as the energy a charge-field system gains or losses when a charge of magnitude e is moved through a potential difference of 1 V. The fundamental charge " e " is equal to $1.6 \times 10^{-19} \text{ C}$, the electron volt is related to the joule as follows:

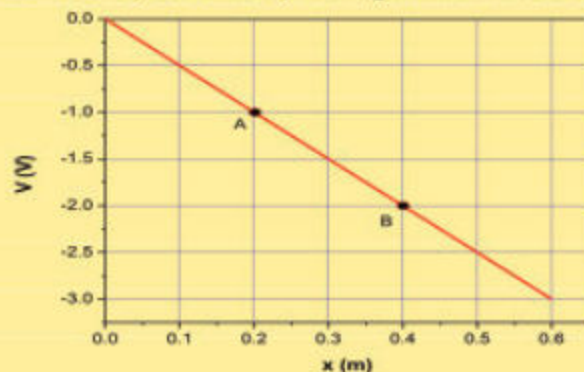
$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ C V}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ C} \left(\frac{\text{J}}{\text{C}} \right)$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Worked Example 8.9

An electron is placed in an xy plane where the electric potential depends on x and y as shown in figure. The scale of the vertical axis is set by one square box equal to 500V. Find the electric field (both cases) acting on the electron between two points as shown in the graph?



$$\frac{GmM_e}{r_1 r_2} (r_2 - r_1) = GmM_e \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

Solution:

Step:1 Write down the known quantities and quantities to be found.

From the graph (a) of the given figure

$$\Delta V = -1000 - (-500) = -500 \text{ V}$$

$$\Delta x = 0.4 - 0.2 = 0.2 \text{ m}$$

From the graph (b)

$$\Delta V = 750 - 250 = 500 \text{ V}$$

$$\Delta x = 0.6 - 0.2 = 0.4 \text{ m}$$

Step:2 Write down the formula and rearrange if necessary.

$$E_x = -\frac{\Delta V}{\Delta x} \text{ and } E_y = -\frac{\Delta V}{\Delta y}$$

Step:3 Put the values in formula and calculate.

$$E_x = -\left(\frac{-500 \text{ V}}{0.2 \text{ m}}\right) = 2500 \text{ V/m}$$

Similarly, from the graph (b)

$$E_y = -\frac{\Delta V}{\Delta y} = -\left(\frac{500 \text{ V}}{0.4 \text{ m}}\right) = -1250 \text{ V/m}$$

Hence, the value of electric field acting on the electron in both cases are 2500 V/m and -1250 V/m.

Self-Assessment Questions:

1. Discuss how potential difference and electric field strength are related.
2. If a proton is released from rest in an electric field, will it move in the direction of increasing or decreasing potential? Explain why.



SUMMARY

- The electrostatic force is an attractive or repulsive force between the charged particles. The electric force between stationary charged body is conventionally known as the electrostatic force. It is also referred to as Coulomb's force.
- Coulomb's Law describes the electrostatic force (electric force) between two charged particles. $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$

where $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N m}^2$ is permittivity of free space. The ratio $\frac{1}{4\pi\epsilon_0}$ is often replaced with Coulomb's constant $k = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2$.

- The **electric field** $\vec{E}$ at some point in space is defined as the electric force $\vec{F}$ that acts on a small positive test charge placed at that point divided by the magnitude q_0 of the test charge. $\vec{E} = \frac{\vec{F}}{q_0}$

- The magnitude of **electric field** set up by a point charge q at a distance r from the charge is,

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

- An **electric dipole** is a pair of equal and opposite charges $+q$ and $-q$ separated by some distance d . Its dipole moment vector $\mathbf{p}$ has magnitude qd and is in the direction of the dipole axis from $-q$ to $+q$. The magnitude of the electric field set up by the dipole at a distant point perpendicular to the dipole axis is $E = \frac{1}{4\pi\epsilon_0} \frac{p}{y^3}$

- The **electric flux** Φ_e of the electric field $\vec{E}$ through vector area $\vec{A}$ is given by $\Phi_e = \vec{E} \cdot \vec{A}$

- The electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ time the total charge enclosed in that surface.

- The **electric potential** is defined as the work done on a charge to move it from lower potential to the higher potential and given as,

$$V = \frac{W}{q_0} = \frac{U}{q_0}$$

- The **electric potential** due to a point charge at any distance r from the charge is

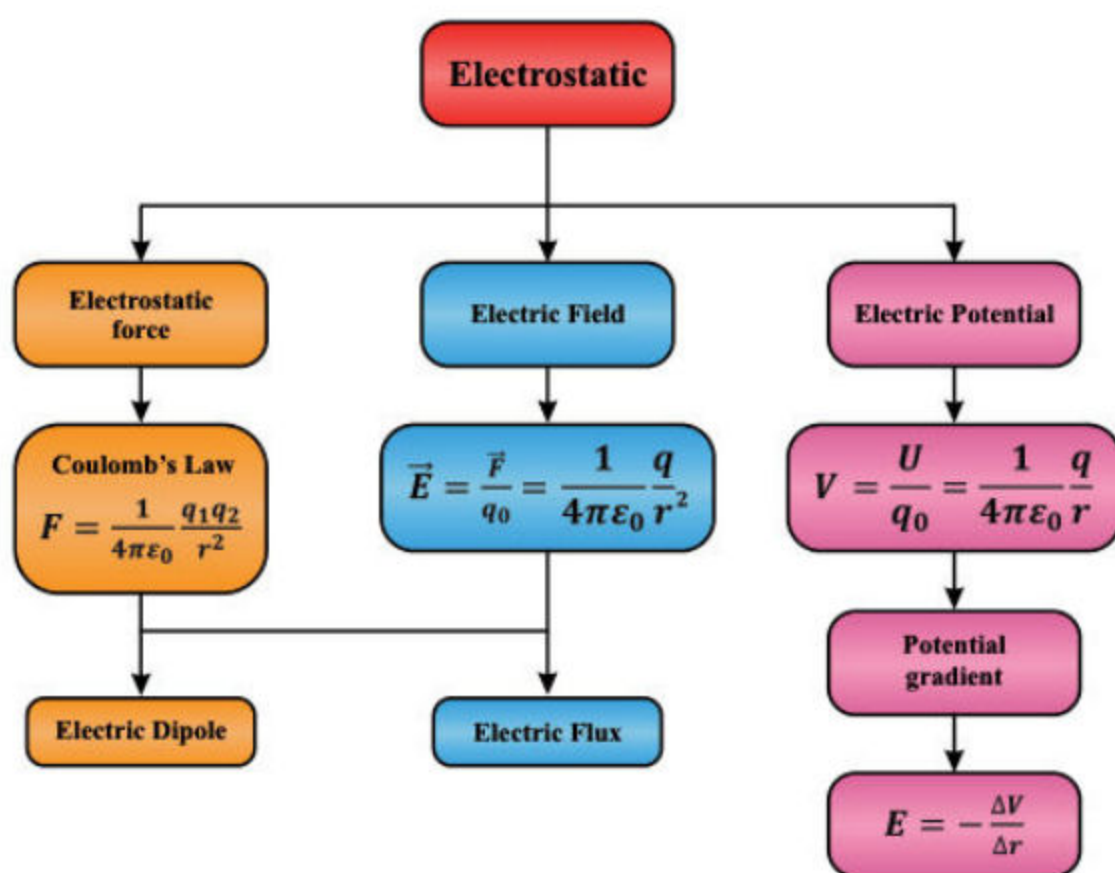
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

- The electric potential associated with a group of point charges is obtained by summing the potentials due to the individual charges.

- The electric field intensity can be calculated from the negative gradient of potential as given below: $E = -\frac{\Delta V}{\Delta r}$

- One electron volt is equivalent to the energy an electron acquires when accelerated across an electric potential difference of one volt. Mathematically, it is represented as:

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ joules}$$





EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. A $2\mu\text{C}$ point charge is located a distance "d" away from $6\mu\text{C}$ point charge, what is the ratio of F_{12}/F_{21} ?
 (a) $1/3$ (b) 3
 (c) 1 (d) 12
2. The minimum charge on an object cannot be less than:
 (a) $1.6 \times 10^{-19}\text{C}$ (b) $3.2 \times 10^{-19}\text{C}$
 (c) 9.1×10^9 (d) No definite value exist
3. Two charges are placed at a certain distance. If the magnitude of each charge is doubled the force will become
 (a) $1/4$ th of its original value (b) 4 times of its original value
 (c) $1/8$ th of its original value (d) 8 times of its original value
4. Which of the following can be deflected while moving in the electric field?
 (a) neutron (b) photon
 (c) electron (d) (a) and (b)
5. The flux through a flat surface of area "A" in a uniform electric field "E" is maximum when the surface area is:
 (a) Parallel to E (b) perpendicular to E
 (c) placed 45° to E (d) placed 60° to E
6. The product of charge "q" and small separation "d" between two charges of same magnitude and opposite in nature is known as:
 (a) Electric dipole (b) Moment arm
 (c) Electric dipole moment (d) Flux of electric field
7. 12 J of work is to be done against an existence electric field to take a charge of 0.01 C from one-point A to another point B. The potential difference between B and A is
 (a) 120 V (b) 1200 V
 (c) 1.2 V (d) 12 V
8. The force between two charges placed in air is F. if air is replaced by a medium of relative permittivity ϵ_r then force is reduced to:
 (a) $F\epsilon_r$ (b) F/ϵ_r
 (c) ϵ_r/F (d) $\epsilon\epsilon_r$
9. The negative gradient of the potential is:
 (a) potential energy (b) voltage
 (c) electric field intensity (d) electric flux
10. The electric flux through a plane area will be half of its maximum value when area is held at angle of _____ with electric field
 (a) 30° (b) 60°
 (c) 45° (d) 90°

Section (B): Structured Questions

CRQs:

1. Why do most objects tend to contain nearly equal numbers of positive and negative charges?
2. When measuring an electric field, could we use a negative rather than a positive test charge?
3. During fair weather, the electric field due to the net charge on Earth points downward. Is Earth charged positively or negatively?
4. How the electric flux through a closed surface is independent on the size or shape of the surface enclosed the charge.
5. What is electric dipole and electric dipole moment?
6. A charged particle is seen to be moving in an electric field along a straight line. How it effects the path of motion of the particle?
7. An electron has a speed of 10^6 m/s. Find its energy in electron volt. (Ans. 2.84 eV)

ERQs:

1. Define electric charge. Discuss the fundamental properties of electric charge, including the principles of conservation of charge and quantization of charge.
2. State and explain Coulombs law. Apply it to calculate the electric field due to an isolated point charge.
3. Define electric flux and explain its significance in the study of electromagnetism. How does it relate to electric fields and charged particles?
4. What is electric potential? Derive expression for potential due to an isolated point charge.
5. Define an electric dipole. Derive formula for the electric field due to an electric dipole at a point "P" placed on its axial line.

Numericals:

1. (a) Calculate the value of two equal charges if they repel one another with a force of 0.1 N when situated 50 cm apart in a vacuum.
(b) What would be the size of the charges if they were situated in an insulating liquid whose permittivity was ten times that of a vacuum? (Ans. $1.7 \mu\text{C}$, $5.3 \mu\text{C}$)
2. How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of the gravitational force on a proton at Earth's surface? (mass of proton = 1.67×10^{-27} kg, charge of proton = 1.6×10^{-19} C (Ans. 0.119 m)
3. An electron of charge 1.6×10^{-19} C is situated in a uniform electric field of intensity 1200-volt cm^{-1} . Find the force on it, its acceleration, and the time it takes to travel 2 cm from rest (electronic mass, $m_e = 9.1 \times 10^{-31}$ kg).
(Ans. 1.92×10^{-14} N, $2.12 \times 10^{16} \text{ ms}^{-2}$, 1.37×10^{-9} s).

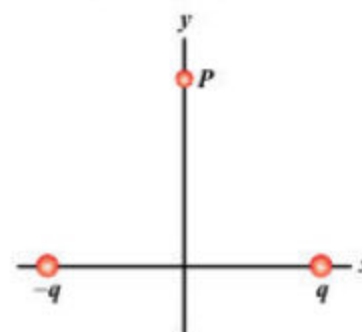
4. An alpha particle (the nucleus of a helium atom) has a mass of 6.64×10^{-27} kg and a charge of $2e$. What are the (a) magnitude and (b) direction of the electric field that will balance the gravitational force on the particle?

(Ans. 2.03×10^7 N/C, the electric field is directed upwards)

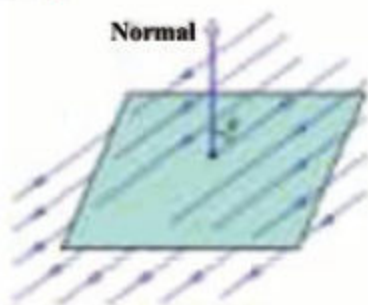
5. A proton and an electron form two corners of an equilateral triangle of side length 2.0×10^{-6} m. What is the magnitude of the net electric field these two particles produce at the third corner? (Ans. 3.6×10^2 N/C)

6. Figure shows two charged particles on an x axis: $-q = -3.20 \times 10^{-19}$ C at $x = -3.00$ m and $q = 3.20 \times 10^{-19}$ C at $x = 3.00$ m. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net electric field produced at point P at $y = 4.00$ m?

(Ans. 1.38×10^{10} N/C, The net electric field points in the $-x$ direction)



7. A proton and an electron form two corners of an equilateral triangle of side length $5 \mu\text{m}$. What is the magnitude of the net electric field these two particles produce at the third corner?
8. The square surface shown in figure measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude $E = 1800$ N/C and with field lines at an angle of $\theta = 35^\circ$ with a normal to the surface, as shown. Take that normal to be directed "outward," as though the surface were one face of a box. Calculate the electric flux through the surface. (Ans. -1.5×10^{-2} N.m²/C)



9. An electron is liberated from the lower of two large parallel metal plates separated by a distance $h = 2$ cm. The upper plate has a potential of 2400 volts relative to the lower. How long does the electron take to reach it? (Ans. 1.2×10^5 Vm⁻¹)
10. Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then 5.0 V, what is the electric field in the region between the plates? (Ans. 6.7×10^2 V/m)

Unit

9

Capacitors

Teaching Periods 5

Weightage % 3



Various types of Capacitors

In this unit student should be able to:

- Explain capacitors as Charge storing Devices.
- Identify types of capacitors used in different field.
- Identify factors affecting the capacitance of a parallel plate capacitor and use equations
 $\epsilon_r = C/C_0$; $C = \epsilon_0 \epsilon_r A/d$.
- Calculate combined capacitance of capacitors in series and in parallel.
- Demonstrate charging and discharging of a capacitor through a resistance.
- Prove that energy stored in capacitor is:
 $W = \frac{1}{2} QV$ and hence $E = \frac{1}{2} CV^2$

9.1.1 Capacitor:

A capacitor is an electrical device which is used to store electrical energy in the same way as the bucket is used for storing water or a tank for storing gas. Each of these devices has fixed capacity which does not depend on the quantity to be stored.

Capacitors are electronic components designed to store electric charge. Figure 9.1 shows the schematic diagram of parallel plate capacitors. They consist of two conductive plates separated by an insulating material called a dielectric. When a voltage is applied across the plates, one plate accumulates a positive charge while the other accumulates an equal and opposite negative charge, resulting in the storage of electric charge in the capacitor.

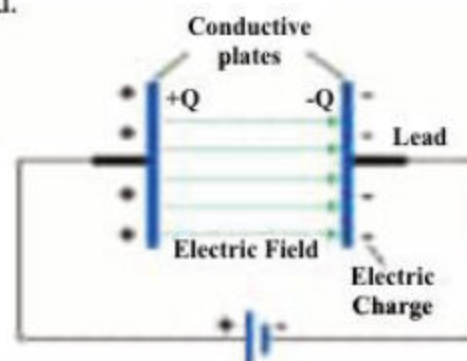


Fig: 9.1
A Parallel Plate Capacitor.

The ability of a capacitor to store charge is determined by its capacitance (C), which is measured in farads (F). The capacitance depends on the physical characteristics of the capacitor, such as the area of the plates, the distance between them, and the properties of the dielectric material.

The quantity of charge Q on a capacitor is directly proportional to the potential difference V between the plates; that is, $Q \propto V$.

$$Q = CV \quad (9.1)$$

Where:

Q = Electric charge stored in the capacitor (measured in coulombs, C)

C = Capacitance of the capacitor (measured in farads, F)

V = Voltage applied across the capacitor (measured in volts, V)

The proportionality constant depends on the shape and separation of Plates.

The capacitance of a capacitor is defined as

“The amount of charge required to create a unit potential difference between Parallel plates.”

From equation (9.1), the capacitance C of a capacitor can be measured as

$$C = \frac{Q}{V}$$

Hence, capacitance is the charge stored per unit potential difference.

DO YOU KNOW?

Capacitors are very useful to reduce voltage fluctuations in electronic power supplies, to transmit signals, to detect electro-magnetic oscillations at radio frequencies and to other important electronic circuits.

DO YOU KNOW?



Capacitor also used as sensor

9.1.2 Types of Capacitor:

Capacitors play a very important role in various electrical circuits, especially radio circuit. Here we classify some common types of capacitors.

Variable Capacitors:

This type of capacitor is used to control the capacitance continuously for tuning transmitters and receivers signals.



Variable capacitor

Film Capacitors:

Film Capacitors consisting of a relatively large family of capacitors with the difference being in their dielectric properties. They are available in almost any value and voltages as high as 1500 volts. Film capacitors are used in power electronics devices, phase shifters, X-ray flashes and pulsed lasers.



Film capacitor

Ceramic capacitors:

These capacitors are used in high-frequency circuits ranging from audio to radio frequency. These capacitors are also called disc capacitors.



Ceramic capacitor

Electrolytic Capacitors:

These capacitors are generally used when very large capacitance values are required. The majority of electrolytic capacitors are polarized. Electrolytic Capacitors are generally used in DC power supply circuits due to their large capacitance's and small size which help to reduce the ripple voltage.



Electrolytic capacitor

9.1.3 Capacitance of a parallel plate Capacitor:

It consists of two parallel metallic plates of equal area A which are separated by a distance d as shown in **figure 9.2**. Both parallel plates carry charges of magnitude $+Q$ and $-Q$ respectively. The surface charge density (σ) is defined as the charge stored (Q) per unit area (A).

$$\sigma = Q / A$$

Therefore, the electric field in this case will be

$$E = \frac{Q}{A\epsilon_0} = \frac{\sigma}{\epsilon_0} \quad \dots \dots (9.2)$$

As the field between the plates is uniform, therefore, the magnitude of the potential difference between the plates equals Ed . i.e., $V=Ed$, Substituting the value of V from equation (9.1), we get,

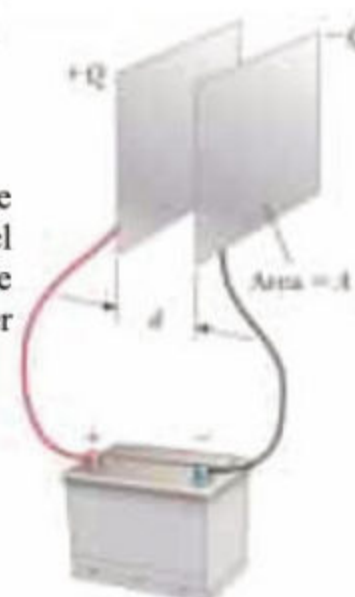


Fig: 9.2

A parallel-plate capacitor consists of two parallel conducting plates, each of area A , separated by a distance d .

$$E = \frac{V}{d} = \frac{Q}{Cd} \quad \dots \dots (9.3)$$

By comparing the equations (9.2) and (9.3), we get relation for capacitance as given below

$$C = \frac{\epsilon_0 A}{d} \quad \dots \dots (9.4)$$

Equation (9.4) shows that the capacitance of a parallel-plate capacitor is proportional to the area of plates and inversely proportional to the plate separation.

Effects of dielectric materials:

Experimentally it is found that capacitance of a capacitor can be increased by filling the space between the plates with dielectrics. Dielectrics are electrically insulating materials that increase the ability of storing charges on the plates of the capacitor. Polythene and waxed paper are the examples of dielectrics.

Microscopically, the molecules of the dielectric materials become polarized by the charged plates of the capacitor as shown in figure 9.3. This polarization is responsible to increase the capacitance of the capacitor. When the dielectric is inserted between the plates of a capacitor, then permittivity of such capacitor will increase by the amount of $\epsilon_0 \epsilon_r$, where ϵ_r is the relative permittivity of a substance and it is called the dielectric constant.

Therefore, the capacitance of such capacitor is given as

$$C_o = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C_o = \epsilon_r \left(\frac{\epsilon_0 A}{d} \right) = \epsilon_r C$$

The above relation shows that the charge storing capacity of a capacitor is enhanced by the dielectric which permits it to store ϵ_r times more charge for the same potential difference.

$$\epsilon_r = \frac{C_o}{C} \quad \dots \dots (9.5)$$

The relative permittivity ϵ_r of the material is the ratio of the capacitance of a capacitor with a given materials filling the space between the plates to the capacitance of the same capacitor when space is evacuated. Typical values of ϵ_r are presented in table 9.1.

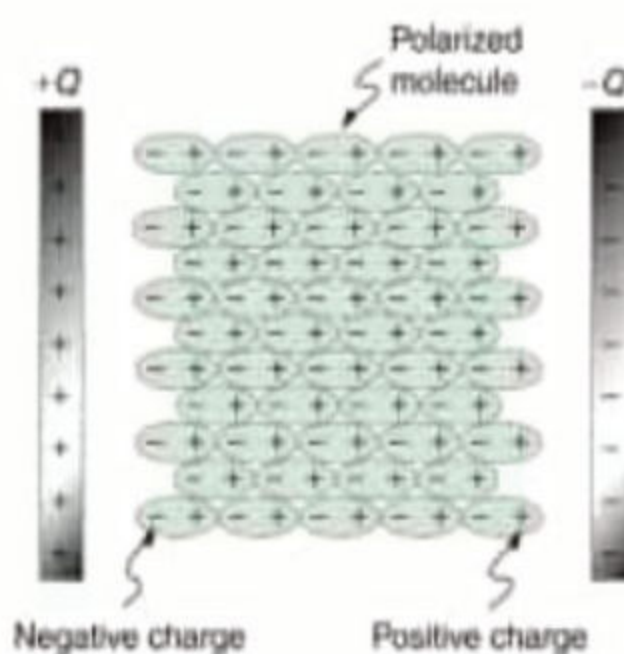


Fig: 9.3
A capacitor filled with dielectric material.

Table No. 9.1: Dielectric Constant (ϵ_r)

S/No.	Insulating material	Dielectric constant
1	Vacuum	1
2	Air	1.0006
3	Teflon	2
4	Mineral Oil	2.2
5	Polyethylene	2.3
6	Waxed paper	2.5
7	Epoxy	3.3
8	PVC	3.7
9	Nylon	4.1
10	Bakelite	5
11	Mica	7
12	Water	81

Worked Example 9.1

A parallel plate capacitor consists of two plates with an area of 0.01 m^2 each, separated by a distance of 0.002 m . The capacitor is filled with a dielectric material having a relative permittivity (ϵ_r) of 4. Calculate the capacitance of this capacitor.

Solution:

Step 1: Formula:

$$C = \epsilon_0 \epsilon_r A / d$$

where:

C is the capacitance,

ϵ_0 is the vacuum permittivity constant (approximately $8.854 \times 10^{-12} \text{ F/m}$),

ϵ_r is the relative permittivity (given as 4),

A is the area of one plate (0.01 m^2),

d is the separation distance between the plates (0.002 m).

Step 2:

Now, substitute the given values into the formula:

$$C = (8.854 \times 10^{-12} \text{ F/m}) \times 4 \times (0.01 \text{ m}^2) / (0.002 \text{ m})$$

$$C \approx 8.854 \times 10^{-12} \times 4 \times (0.01 / 0.002) \text{ F}$$

$$C \approx 3.5416 \times 10^{-11} \text{ F}$$

So, the capacitance of the parallel plate capacitor is approximately $3.5416 \times 10^{-11} \text{ Farads (F)}$.

Self-Assessment Questions:

1. The capacitance of a capacitor formed by two parallel metal plates each 200 cm^2 in area separated by a dielectric 4 mm thick is $0.0004 \text{ microfarads}$. A potential difference of $20,000 \text{ V}$ is applied. Calculate the total charge on the plates?

9.1.4 Combinations of Capacitors:

Two or more capacitors often are combined in electric circuits. We can calculate the equivalent capacitance of certain combinations with the help of a circuit diagram. The circuit symbols for capacitors, batteries, and switches are shown in figure 9.4.

Parallel Combination of capacitors:

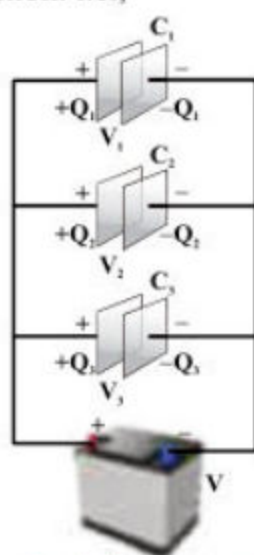
In a parallel combination of capacitors, two or more capacitors are connected side by side with their positive terminals connected together and their negative terminals connected together. This arrangement allows the capacitors to share the same voltage across their terminals while increasing the total capacitance of the combination.

When capacitors are connected in parallel, the total capacitance (C_{total}) of the combination is the sum of the individual capacitances (C_1, C_2, C_3 , and so on) of the capacitors connected:

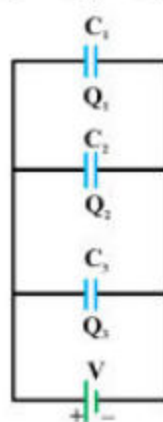
$$C_{\text{total}} = C_1 + C_2 + C_3 + \dots + C_n$$

Consider three capacitors are connected in parallel combination as shown figure 9.5 and all three diagrams are equivalent. we know that potential difference across each capacitors connected in parallel are the same and equal to the potential difference applied across the combination. i.e.,

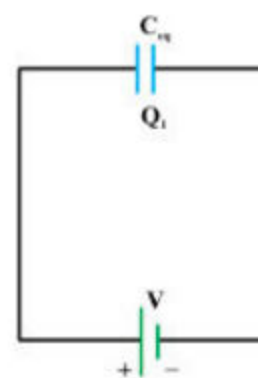
$$V = V_1 = V_2 = V_3$$



(a) a pictorial representation of three capacitors connected in parallel to a battery



(b) circuit diagram showing the three capacitors connected in parallel to a battery



(c) circuit diagram showing the equivalent capacitance of the capacitors connected in parallel.

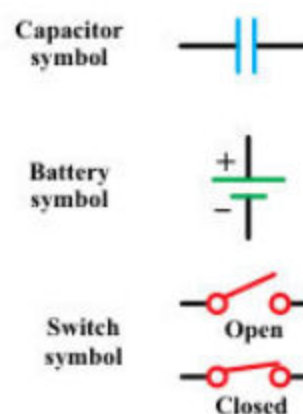


Fig: 9.4
Circuit symbols for capacitors, batteries, and switches.

Unit 9: Capacitors

In general, when a potential difference V is applied across several capacitors connected in parallel, that potential difference V is same across each capacitor. The total charge Q stored on the equivalent capacitor is the sum of the charges stored on all the capacitors.

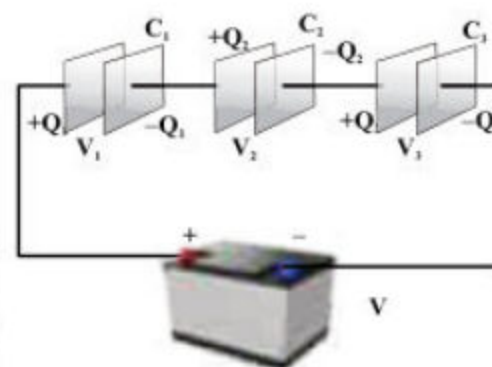
To find an expression for equivalent capacitance C_{eq} , use the eq. (9.1) to find the charge on each actual capacitor and equivalent capacitor.

$$Q_1 = C_1 V, \quad Q_2 = C_2 V, \quad Q_3 = C_3 V \quad \text{and} \quad Q = C_{eq} V$$

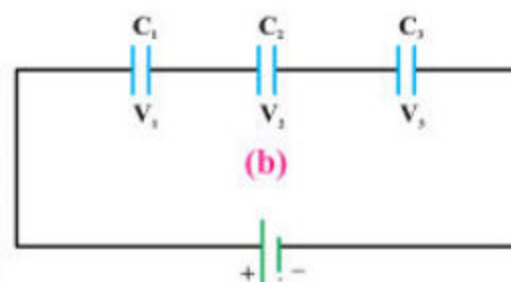
The total charge on the parallel combination is given as

$$\begin{aligned} Q &= Q_1 + Q_2 + Q_3 \\ C_{eq} V &= C_1 V + C_2 V + C_3 V \\ C_{eq} &= C_1 + C_2 + C_3 \quad \dots (9.6) \end{aligned}$$

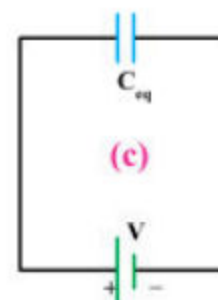
Therefore, the equivalent capacitance of a parallel combination of capacitors is the algebraic sum of the individual capacitances and has greater value than any of the individual capacitances. This combination is used in the circuit where the high capacitance is required because equivalent capacitor store more energy.



(a)



(b)



(c)

Fig: 9.6

(a) a pictorial representation of two capacitors connected in series to a battery

(b) circuit diagram showing the two capacitors connected in series to a battery

(c) circuit diagram showing the equivalent capacitance of the capacitors connected in series

Series Combination of capacitors:

In a series combination of capacitors, two or more capacitors are connected one after the other in a single line, with the positive terminal of one capacitor connected to the negative terminal of the next capacitor. This arrangement allows the capacitors to share the same charge while effectively reducing the total capacitance of the combination compared to individual capacitors.

Three capacitors are connected in series combination as shown Figure 9.6 (a, b, c) and all three diagrams are equivalent. In series means that the capacitors are connected by a wire and there is no other way of flowing of charge. When a potential V is applied across them, the same amount Q of charge will appear on each of the capacitor. The battery directly produces charges on only the two plates to which it is connected. Charges that are produced on the other plates are due to the simple impact of electric field induction. Due to this the charges can be distributed over all other plates connected in the circuit. Therefore, the charges on capacitors connected in series are the same.

In general, the total potential difference across any number of capacitors connected in series is the sum of the potential differences across the individual capacitors.